Analytic Test of RT-GSFit

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Flux convention

Throughout this document, ψ is total poloidal flux in weber, following COCOS 13, not flux per radian. The physical permeability μ_0 contains no factor of 2π. With $F = RB_\phi$ and primes denoting derivatives with respect to this total flux,

$$B_R = -\frac{1}{2\pi R}\frac{\partial \psi}{\partial z}, \qquad B_z = \frac{1}{2\pi R}\frac{\partial \psi}{\partial R}, \qquad \Delta^* \psi = -2\pi\mu_0 R j_\phi.$$

The earlier per-radian version of this derivation used $\psi_{\mathrm{rad}} = \psi/(2\pi)$. Rescaling flux does not change the magnetic field, current density or normalised flux.

1 Deriving Bessel's Eqn from the Grad-Shafranov Eqn

The Grad-Shafranov equation is given by

$$\frac{\partial^2 \psi}{\partial R^2} - \frac{1}{R}\frac{\partial \psi}{\partial R} + \frac{\partial^2 \psi}{\partial z^2} = -(2\pi)^2 \left[\mu_0 R^2 p'(\psi) + F(\psi)F'(\psi)\right].$$

The squared factor follows because both flux derivatives and the left-hand-side flux change when converting from flux per radian to total flux. We will assume a plasma beta, $\beta = 0$. We are interested in solving the Grad-Shafranov equation inside the LCFS. We take the large aspect-ratio limit $\rho_b/R_o \ll 1$, neglecting the $R^{-1}\partial_R\psi$ term relative to the second derivatives. Hence, we can simplify the Grad-Shafranov to

$$\frac{\partial^2 \psi}{\partial R^2} + \frac{\partial^2 \psi}{\partial z^2} = -(2\pi)^2 F(\psi)F'(\psi) = -2\pi\mu_0 R j_\phi.$$

We will assume poloidal symmetry, and so we assume $\psi = \psi(\rho)$. Let

$$\rho = \sqrt{(R-R_o)^2 + (z-z_o)^2}.$$
$$R - R_o = \rho\cos(\theta),$$
$$z - z_o = \rho\sin(\theta),$$
$$\hat{\rho} = \cos(\theta)\hat{\mathbf{R}} + \sin(\theta)\hat{\mathbf{z}},$$
$$\hat{\theta} = -\sin(\theta)\hat{\mathbf{R}} + \cos(\theta)\hat{\mathbf{z}}.$$

Note that

$$\begin{aligned}
\frac{\partial \psi}{\partial R} &= \frac{\partial \rho}{\partial R}\frac{d\psi}{d\rho} \\
&= \frac{(R-R_o)}{\sqrt{(R-R_o)^2 + (z-z_o)^2}}\frac{d\psi}{d\rho} \\
&= \cos(\theta)\frac{d\psi}{d\rho}, \\
\frac{\partial \psi}{\partial z} &= \frac{\partial \rho}{\partial z}\frac{d\psi}{d\rho} \\
&= \frac{(z-z_o)}{\sqrt{(R-R_o)^2 + (z-z_o)^2}}\frac{d\psi}{d\rho} \\
&= \sin(\theta)\frac{d\psi}{d\rho},
\end{aligned}$$

$$\begin{aligned}
\frac{\partial\psi}{\partial R}\hat{\mathbf{R}} + \frac{\partial\psi}{\partial z}\hat{\mathbf{z}} &= \frac{d\psi}{d\rho}(\cos(\theta)\hat{\mathbf{R}} + \sin(\theta)\hat{\mathbf{z}}) \\
&= \frac{d\psi}{d\rho}\hat{\rho}, \\
\hat{\mathbf{R}} \cdot \frac{\partial\hat{\rho}}{\partial R} &= \frac{\partial}{\partial R}\left(\frac{R - R_o}{\sqrt{(R-R_o)^2 + (z-z_o)^2}}\right) \\
&= \frac{\sin^2\theta}{\rho} \\
\hat{\mathbf{z}} \cdot \frac{\partial\hat{\rho}}{\partial z} &= \frac{\partial}{\partial z}\left(\frac{z - z_o}{\sqrt{(R-R_o)^2 + (z-z_o)^2}}\right) \\
&= \frac{\cos^2\theta}{\rho} \\
\frac{\partial^2\psi}{\partial R^2} + \frac{\partial^2\psi}{\partial z^2} &= \left(\hat{\mathbf{R}}\frac{\partial}{\partial R} + \hat{\mathbf{z}}\frac{\partial}{\partial z}\right) \cdot \left(\frac{\partial\psi}{\partial R}\hat{\mathbf{R}} + \frac{\partial\psi}{\partial z}\hat{\mathbf{z}}\right) \\
&= \left(\hat{\mathbf{R}}\frac{\partial}{\partial R} + \hat{\mathbf{z}}\frac{\partial}{\partial z}\right) \cdot \left(\frac{d\psi}{d\rho}\hat{\rho}\right) \\
&= \cos(\theta)\frac{\partial}{\partial R}\left(\frac{d\psi}{d\rho}\right) + \frac{d\psi}{d\rho}\hat{\mathbf{R}} \cdot \frac{\partial\hat{\rho}}{\partial R} + \sin(\theta)\frac{\partial}{\partial z}\left(\frac{d\psi}{d\rho}\right) + \frac{d\psi}{d\rho}\hat{\mathbf{z}} \cdot \frac{\partial\hat{\rho}}{\partial z} \\
&= \frac{d^2\psi}{d\rho^2} + \frac{1}{\rho}\frac{d\psi}{d\rho}.
\end{aligned}$$

Hence, we can rewrite the Grad-Shafranov equation as

$$\frac{d^2\psi}{d\rho^2} + \frac{1}{\rho}\frac{d\psi}{d\rho} = -(2\pi)^2 F(\psi)F'(\psi).$$

For this analytic case, choose

$$(2\pi)^2 FF' = a(1 - \psi_N),$$

where a is the constant amplitude of the Laplacian source, not a fitted coefficient, and

$$\psi_N = \frac{\psi_o - \psi}{\psi_o - \psi_b},$$

where ψ_b is ψ at the last closed flux surface and ψ_o is the value of ψ at the o-point. Let

$$\psi(\rho) = \psi'(\rho) + \psi_b,$$

$$a' = \frac{a}{(\psi_o - \psi_b)},$$

hence by substituting our expression for ψ in terms of ψ' and a in terms of a' we get

$$\frac{d^2\psi'}{d\rho^2} + \frac{1}{\rho}\frac{d\psi'}{d\rho} = -a'\psi'.$$

Multiplying through by ρ^2 and rearranging we arrive at an equation that is nearly Bessel's equation of order 0

$$\rho^2\frac{d^2\psi'}{d\rho^2} + \rho\frac{d\psi'}{d\rho} + a'\rho^2\psi' = 0.$$

For the Bessel solution we choose $a' > 0$. Let

$$\begin{aligned}
s &= \sqrt{a'}\rho, \\
\implies \rho &= \frac{s}{\sqrt{a'}}, \\
\implies \frac{\partial}{\partial\rho} &= \sqrt{a'}\frac{\partial}{\partial s},
\end{aligned}$$

$$\implies \frac{\partial^2}{\partial \rho^2} = a' \frac{\partial^2}{\partial s^2},$$

$$\implies \rho^2 = \frac{s^2}{a'}$$

Hence, the Grad-Shafranov equation becomes

$$s^2 \frac{\partial^2 \psi'}{\partial s^2} + s \frac{\partial \psi'}{\partial s} + s^2 \psi' = 0,$$

which is exactly of the same form as Bessel's equation of order 0.

2 Solving Bessels' equation

We know that ψ' is of the form

$$\psi' = C_1 J_0(s) + C_2 Y_0(s),$$

where J_0 and Y_0 are Bessel function of order zero of the first and second kind respectively. We know that ψ is finite at $R = R_o$, $z = z_o$, hence, $C_2 = 0$. Past the last closed flux surface, there is no plasma current. Hence, the full solution for ψ' is

$$\psi'(\rho) = C_1 J_0(\sqrt{a'}\rho).$$

3 Tidying up solution

We know that

$$\psi = C_1 J_0(\sqrt{a'}\rho) + \psi_b,$$

which we can rewrite as

$$\psi = C_1 J_0\left(a^* \frac{\rho}{\rho_b}\right) + \psi_b,$$

where

$$a^* = \sqrt{\frac{a\rho_b^2}{\psi_o - \psi_b}}.$$

We know that $\psi(\rho_b) = \psi_b$, hence,

$$C_1 J_0(a^*) = 0.$$

We will assume that we have chosen a such that a^* is a zero of J_0. We know that $\psi(0) = \psi_o$, hence

$$\psi = (\psi_o - \psi_b) J_0\left(a^* \frac{\rho}{\rho_b}\right) + \psi_b,$$

4 Solution for $\rho > \rho_b$

For $\rho > \rho_b$ there is no plasma current and the Grad-Shafranov equation is given by

$$\rho^2 \frac{d^2\psi}{d\rho^2} + \rho \frac{d\psi}{d\rho} = 0$$

The equation has the solution

$$\psi = C_1 \ln\left(\frac{\rho}{\rho_b}\right) + \psi_b,$$

and satisfies continuity at $\rho = \rho_b$. We also require continuity of the derivative of ψ, note that

$$\frac{d}{d\rho}\left[(\psi_o - \psi_b) J_0\left(a^* \frac{\rho}{\rho_b}\right)\right] = -\frac{(\psi_o - \psi_b)a^*}{\rho_b} J_1\left(a^* \frac{\rho}{\rho_b}\right),$$

$$\frac{d}{d\rho} C_1 \ln\left(\frac{\rho}{\rho_b}\right) = \frac{C_1}{\rho},$$

hence,

$$\frac{C_1}{\rho_b} = -\frac{(\psi_o - \psi_b)a^*}{\rho_b} J_1(a^*),$$
$$\implies C_1 = -(\psi_o - \psi_b)a^* J_1(a^*),$$

Hence, for $\rho \geq \rho_b$

$$\psi = -(\psi_o - \psi_b)a^* J_1(a^*) \ln\left(\frac{\rho}{\rho_b}\right) + \psi_b.$$

5 Total current

For $\rho > \rho_b$, the poloidal field is given by

$$B_\theta = \frac{1}{2\pi R}\frac{d\psi}{d\rho},$$
$$= -\frac{(\psi_o - \psi_b)a^* J_1(a^*)}{2\pi(R_o + \rho\cos\theta)\rho}.$$

The direction of increasing θ has normal $-\hat{\phi}$. Thus Ampere's law gives a negative circulation for positive toroidal current. For a contour $\rho = \rho_0 > \rho_b$ with $\rho_0 < R_o$,

$$\oint_{\rho=\rho_0} B_\theta \rho_0 d\theta = -\frac{(\psi_o - \psi_b)a^* J_1(a^*)}{R_o\sqrt{1 - \rho_0^2/R_o^2}}$$
$$\simeq -\frac{(\psi_o - \psi_b)a^* J_1(a^*)}{R_o} = -\mu_0 I_{\text{total}}.$$

Taking the leading-order term, we know that

$$I_{\text{total}} = \frac{(\psi_o - \psi_b)a^* J_1(a^*)}{R_o\mu_0},$$

$$\implies \psi_o - \psi_b = \frac{\mu_0 I_{\text{total}} R_o}{a^* J_1(a^*)}.$$

Here I_{total} is the prescribed current parameter (called `ip` in the code). Its identification with integrated current is leading order in ρ_b/R_o, consistent with the Laplacian approximation; it is not an exact finite-aspect-ratio equilibrium.

6 Calculating expressions for ψ_o, ψ_b

We need to ensure that ψ is zero at $(R, z) = (0, 0)$. Where we assume this is outside the region $\rho < \rho_b$. Let

$$d = \sqrt{R_o^2 + z_o^2}.$$

We impose $\psi(d) = 0$. Hence

$$\psi_b = \mu_0 I_{\text{total}} R_o \ln\left(\frac{d}{\rho_b}\right),$$

$$\implies \psi_o = \mu_0 I_{\text{total}} R_o \left[\ln\left(\frac{d}{\rho_b}\right) + \frac{1}{a^* J_1(a^*)}\right].$$

The zero at the origin fixes an additive flux gauge; the large-aspect-ratio field approximation is used near the plasma, not near $R = 0$.

7 Full solution

The full solution for ψ is given by

$$\psi(\rho) = \begin{cases} \Delta_\psi J_0\left(a^* \dfrac{\rho}{\rho_b}\right) + \psi_b, & \rho \leq \rho_b, \\ -\Delta_\psi a^* J_1(a^*) \ln\left(\dfrac{\rho}{\rho_b}\right) + \psi_b, & \rho \geq \rho_b, \end{cases}$$

where

$$\begin{aligned} \Delta_\psi &= \psi_o - \psi_b \\ \psi_o &= \mu_0 I_{\text{total}} R_o \left[\ln\left(\frac{d}{\rho_b}\right) + \frac{1}{a^* J_1(a^*)}\right], \\ \psi_b &= \mu_0 I_{\text{total}} R_o \ln\left(\frac{d}{\rho_b}\right), \\ \Delta_\psi &= \frac{\mu_0 I_{\text{total}} R_o}{a^* J_1(a^*)}, \\ a &= \frac{(a^*)^2(\psi_o - \psi_b)}{\rho_b^2} \\ &= \frac{\mu_0 a^* I_{\text{total}} R_o}{J_1(a^*)\rho_b^2}, \end{aligned}$$

and a^* is the coordinate of a positive zero of J_0. Hence, the solution is not unique but if we impose that ψ needs to decrease monotonically away from ψ_o then this forces a^* to be the first positive zero of J_0.

8 Current density and fitted coefficient

The current profile corresponding to the Laplacian source is

$$j_\phi = \frac{a}{2\pi\mu_0 R}(1 - \psi_N) = \frac{a^* I_{\text{total}} R_o}{2\pi J_1(a^*)\rho_b^2 R}(1 - \psi_N) \quad (\rho \leq \rho_b),$$

and zero outside. RT-GSFit fits the coefficient of the basis $(1 - \psi_N)/R$:

$$c_{\text{ff}} = \frac{a}{2\pi\mu_0} = \frac{a^* I_{\text{total}} R_o}{2\pi J_1(a^*)\rho_b^2}.$$

This is the coefficient labelled `FF_PRIME_MU0` in the implementation. In this label, the historical F' uses the per-radian flux derivative: $c_{\text{ff}} = F(dF/d\psi_{\text{rad}})/\mu_0$. With the total-flux prime convention used in this document, it is instead $c_{\text{ff}} = 2\pi F F'/\mu_0$. The coefficient and current density are unchanged by the flux normalisation change.

9 Magnetic field

The magnetic field is given by

$$\begin{aligned} B_R &= B_\theta \hat{\vec{\theta}} \cdot \hat{\vec{R}} \\ &= -\frac{1}{2\pi R}\sin(\theta)\frac{d\psi}{d\rho} \\ &= -\frac{1}{2\pi R}\frac{z - z_o}{\rho}\frac{d\psi}{d\rho} \\ B_z &= B_\theta \hat{\vec{\theta}} \cdot \hat{\vec{z}} \\ &= \frac{1}{2\pi R}\cos(\theta)\frac{d\psi}{d\rho} \\ &= \frac{1}{2\pi R}\frac{R - R_o}{\rho}\frac{d\psi}{d\rho} \end{aligned}$$